Computer Organisation_2023_PYQ

Group-A (Very Short Answer Type Question)

Answer any ten of the following: [1 x 10 = 10]

1. What is a program counter?
2. Floating point representation is used to store what type of values?
3. Is Excess 3-code a weighted code?
4. $(2FAOC)_{16}$ is equivalent to _( )$_2$_ .
5. A self-contained sequence of instructions that performs a given computational task is called _____.
6. What are the parameters which influence the characteristics of a microprocessor?
7. The addressing mode used in an instruction of the form ADD X, Y, is _____ mode.
8. Simplify the following expression using Boolean algebra -- $AB + A\overline{B}$
9. The following transfer statement specifies a memory. Explain the memory operation: M[AR] ← R3
10. A digital computer has a common bus system for 16 registers of 32 bits each. The bus is constructed with multiplexers. How many selection inputs are there in each multiplexer?
11. How many flip-flops will be complemented in a 10-bit binary counter to reach the next count after 1001100111?
12. Logic X-OR operation of $(4ACO)_H$ & $(B53F)_H$ results _____ in hexadecimal.

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Group-B (Short Answer Type Question)

Answer any three of the following: [5 x 3 = 15]

1. Explain FLYNN Classification with suitable examples.
2. Explain floating point representation technique for IEEE 754 standard.
3. Describe sign extension technique for signed number representation.
4. A machine has a 32-bit architecture, with 1-word long instructions. It has 64 registers, each of which is 32 bits long. It needs to support 45 instructions, which have an

immediate operand in addition to two register operands. Assuming that the immediate operand is an unsigned integer, what is the maximum value of the immediate operand?

5. State the differences between Hardwired and Micro-programmed Control Unit.

Group-C (Long Answer Type Question)

Answer any three of the following: [15 x 3 = 45]

1. (a) Explain 4-bit Ripple Carry adder using Full Adder with a suitable block diagram. [5]
   (b) What is the binary value 0.011010 in decimal? [5]
   (c) What is 0.687510 in binary? [5]
2. (a) Consider a 4-way set associative cache consisting of 128 lines with a line size of 64 words. The CPU generates a 20-bit address of a word in main memory. Find the number of bits in the TAG, LINE and WORD fields respectively. [7]
   (b) Explain Virtual memory with suitable examples. [8]
3. Explain 0-address, 1-address, 2-address & 3-address instruction format with suitable examples. [15]
4. (a) Explain fixed point number system for signed numbers. [9]
   (b) Represent the signed number representation for 4-bit numbers of sign magnitude, 1's complement and 2's complement. [6]

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